

COLLABORATIVE AFFORDANCE NEGOTIATION: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

We study human-robot collaboration. The motivating bet is: Model affordances as negotiated between human intent and robot physical capability.. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For human-robot collaboration, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2019; 2022a; 2025a;b; 2026; 2022b). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

3 MECHANISM

Given state s and action a , the model returns an observed next-state prediction plus a finite atlas $B(s, a)$ of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

Variant	Split	Success	Tail risk
observed_only_wam	nominal	0.537	0.463
observed_only_wam	hidden_mode_shift	0.145	0.855
augmentation_as_data_wam	nominal	0.647	0.353
augmentation_as_data_wam	hidden_mode_shift	0.215	0.785
scalar_uncertainty_wam	nominal	0.685	0.315
scalar_uncertainty_wam	hidden_mode_shift	0.325	0.675
proposed_branch_mechanism	nominal	0.780	0.220
proposed_branch_mechanism	hidden_mode_shift	0.598	0.402

5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

6 CONCLUSION

Collaborative Affordance Negotiation argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

REFERENCES

- Physical human-robot collaboration: Robotic systems, learning methods, collaborative strategies, sensors, and actuators, 2019. <https://doi.org/10.1109/tcyb.2019.2947532>.
- Cooperative transportation with mobile manipulator: A capability map-based framework for physical human-robot collaboration, 2022a. <https://doi.org/10.1109/tmech.2022.3155601>.
- Enabling impedance-based physical human-multi-robot collaboration: Experiments with four torque-controlled manipulators, 2022b. <https://doi.org/10.1177/02783649211053650>.
- Gat-grasp: Gesture-driven affordance transfer for task-aware robotic grasping, 2025a. <http://arxiv.org/abs/2503.06227v1>.
- Transforming monolithic foundation models into embodied multi-agent architectures for human-robot collaboration, 2025b. <http://arxiv.org/abs/2512.00797v1>.
- Perception-decision-execution coordination mechanism driven dynamic autonomous collaboration method for human-like collaborative robot based on multimodal large language model, 2026. <https://doi.org/10.1016/j.rcim.2025.103167>.